Computer Communication and Networks

Presentation & Application Layer

Session 18

Tutor: Karma Wangchuk





Presentation & Application Layer

Learning Objectives: At the end of the lesson, you are expected to be able to:

- Introduction
- Domain Name System
- Internet Protocol Security (IPSec)
- Firewall
- Wireless Security
- Shared Key Authentication



Presentation & Application Layer

Domain Name System

- Domain Name System (DNS) is a critical component of the Internet infrastructure that plays a fundamental role in connecting users to **websites**, **services**, and **resources** across the World Wide Web.
  - It is essentially the "**phone book**" of the internet, translating user-friendly domain names (like www.example.com) into numerical IP addresses (such as 192.0.2.1) that computers and network devices use to locate one another on the internet



Presentation & Application Layer

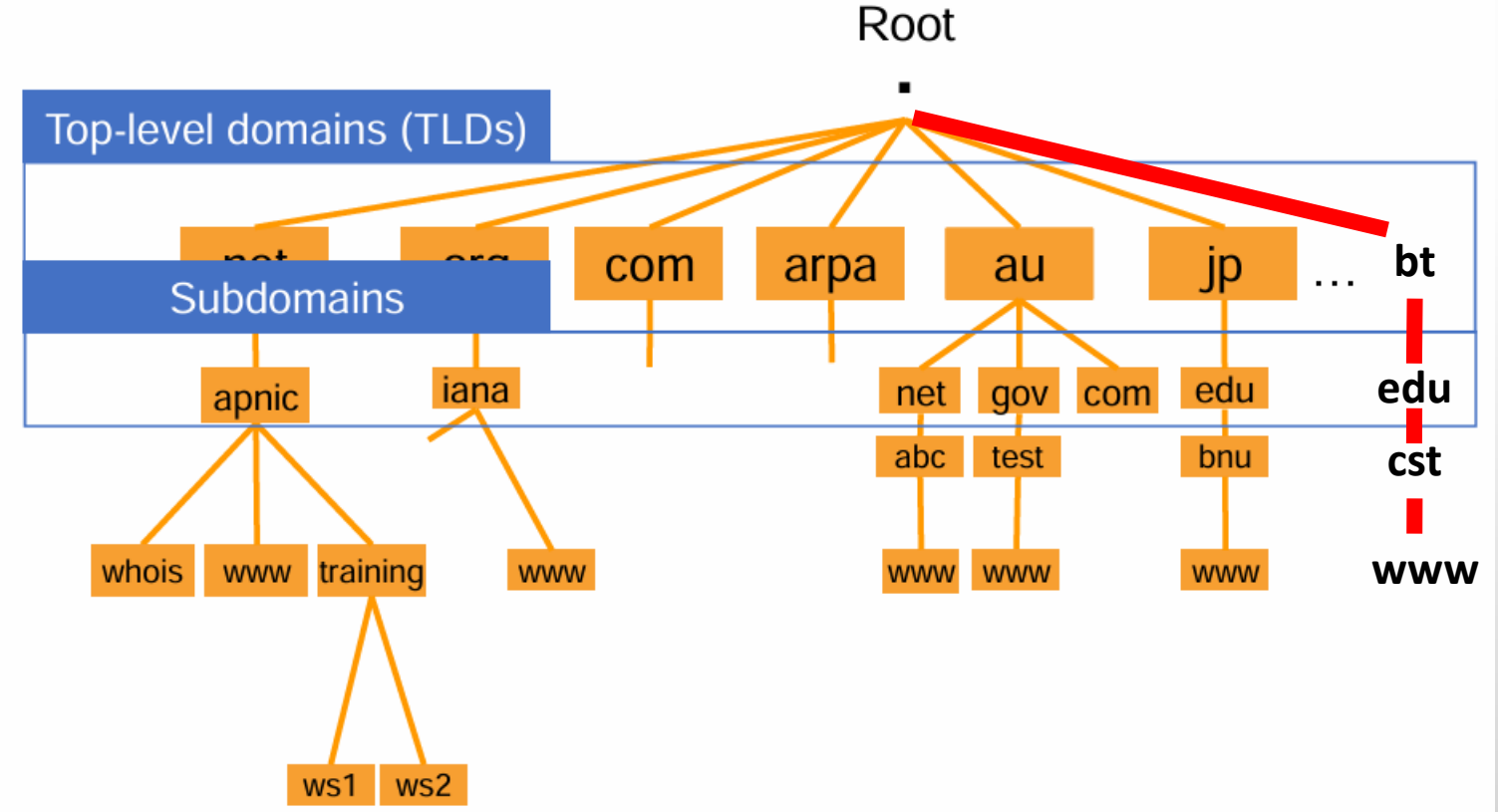
Domain Name System

- DNS is used to resolve human-readable hostnames:
  - Example
    - ✓ www.google.com = 8.8.8.8
    - ✓ www.cst.edu.bt = ??? → nslookup

- When you visit http://www.google.com in a browser, your computer uses DNS to retrieve the website's IP address of 8.8.8.8
- Without DNS, you would only be able to visit the website by visiting its IP address directly.

DNS Hierarchy Tree

- The DNS is organized hierarchically, with a structured naming system to ensure that domain names are unique and globally resolvable.



www.cst.edu.bt



Presentation & Application Layer

DNS Namespace

- Key components of DNS System:
  1. Domain Name
  2. Top-Level Domain
  3. Domain Name Registrar
  4. Authoritative Name Server
  5. Recursive Resolver
  6. Root Name Servers
  7. Caching DNS Servers

- A domain name is a human-readable label that represents a specific location or resource on the internet.
- Domain names are structured as a hierarchy, with levels separated by dots (periods).
- For example: www.google.com has three parts: “www” (subdomain), “google” (second-level domain), and “com” (top-level domain)



Presentation & Application Layer

DNS Namespace

- Key components of DNS System:
  1. Domain Name
  2. Top-Level Domain
  3. Domain Name Registrar
  4. Authoritative Name Server
  5. Recursive Resolver
  6. Root Name Servers
  7. Caching DNS Servers

- TLDs are the highest level in the DNS hierarchy and represent categories of domain names.
- Common examples include ".com," ".org," ".net," and country-code TLDs like ".bt .uk .in and .jp
- ICANN (Internet Corporation for Assigned Names and Numbers) manages the assignment of TLDs.



Presentation & Application Layer

DNS Namespace

- Key components of DNS System:
  1. Domain Name
  2. Top-Level Domain
  3. Domain Name Registrar
  4. Authoritative Name Server
  5. Recursive Resolver
  6. Root Name Servers
  7. Caching DNS Servers

- Registrars are organizations accredited by ICANN to sell domain name registrations.
- They allow individuals and organizations to reserve and manage domain names within specific TLDs.



Presentation & Application Layer

DNS Namespace

- Key components of DNS System:
  1. Domain Name
  2. Top-Level Domain
  3. Domain Name Registrar
  4. Authoritative Name Server
  5. Recursive Resolver
  6. Root Name Servers
  7. Caching DNS Servers

- These are DNS servers that store and manage the DNS records for a specific domain.
- For example, http://ns1.example.com/ might be the authoritative name server for the http://example.com/ domain.



Presentation & Application Layer

DNS Namespace

- Key components of DNS System:
  1. Domain Name
  2. Top-Level Domain
  3. Domain Name Registrar
  4. Authoritative Name Server
  5. Recursive Resolver
  6. Root Name Servers
  7. Caching DNS Servers

Recursive Resolver

- These are DNS servers operated by internet service providers (ISPs) or third-party DNS service providers.
- They are responsible for receiving DNS queries from client devices and recursively resolving domain names by querying authoritative name servers



Presentation & Application Layer

DNS Namespace

- Key components of DNS System:
  1. Domain Name
  2. Top-Level Domain
  3. Domain Name Registrar
  4. Authoritative Name Server
  5. Recursive Resolver
  6. Root Name Servers
  7. Caching DNS Servers

Root Name Servers

- At the top of the DNS hierarchy are 13 root name servers maintained by various organizations worldwide.
- These servers hold information about the TLDs and provide crucial pointers to authoritative name servers for each TLD.



Presentation & Application Layer

DNS Namespace

- Key components of DNS System:
  1. Domain Name
  2. Top-Level Domain
  3. Domain Name Registrar
  4. Authoritative Name Server
  5. Recursive Resolver
  6. Root Name Servers
  7. Caching DNS Servers

Catching DNS Servers

- Provided by internet service providers (ISPs) or used by individuals and organizations.
- Caching DNS servers temporarily store DNS records they've recently looked up.
- When a user queries a domain, these servers check their cache first before querying authoritative DNS servers, which helps reduce DNS query load



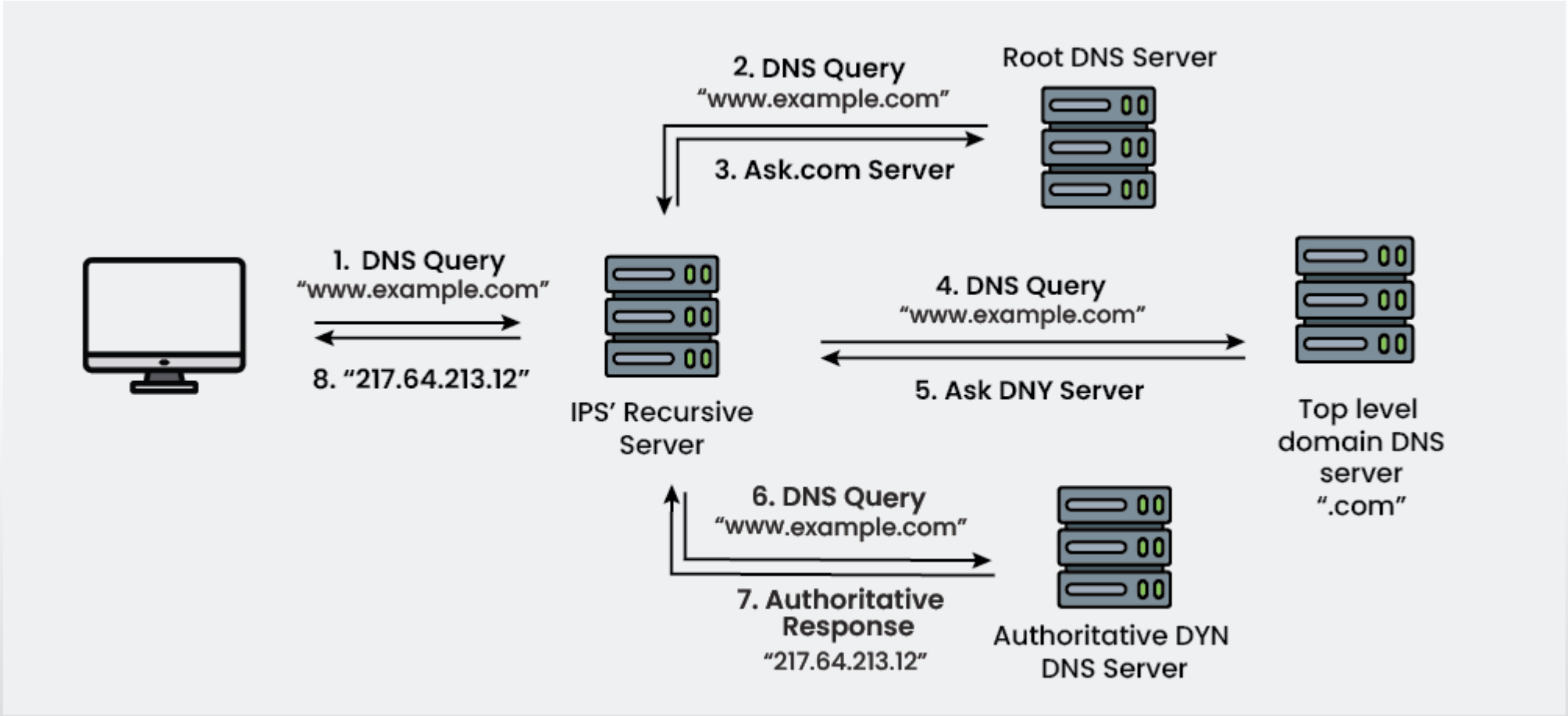
Presentation & Application Layer

- **Domain Name System: working**
  - When you visit a domain such as **facebook.com**
    - Your computer follows a series of steps to turn the human-readable web address into a machine-readable IP address.
  - This happens every time you use a domain name, whether you are viewing websites or sending email



Presentation & Application Layer

How does DNS work?



DNS Image



Presentation & Application Layer

Electronic Mail:

- It is most commonly known as email.
  - ✓ It has been about three decades
- It is faster and cheaper than paper mail.
- It has been a popular application since the early days of the Internet.
  - ✓ Before 1990, it was mostly used in academia.
  - ✓ During 1990s, it became known to the public at large and grew exponentially.
- Other forms of network communication such as instant messaging and VoIP have been expended.
  - ✓ But email remains the workhorse of the Internet.



Presentation & Application Layer

- **World Wide Web**:
  - It is called web in general.
  - It is the an architectural Framework for accessing linked content spread out over millions of machines all over Internet.
  - The web began in 1989 at European Center for Nuclear Research [CERN].
    - The initial idea was to help large team to collaborate using a constantly changing collection of reports, blueprints, photos, and other documents by experiments in particle physics.



Presentation & Application Layer

- **World Wide Web**:
  - The proposal for a web of linked documents came from CERN physicist Tim Berners-Lee
    - ✓ The first prototype was operational 18 months later
      - Prototype was based on text
  - The first graphical browser was developed by Marc Andreessen at University of Illinois.
    - ✓ It was called Mosaic and released in February 1993.



Presentation & Application Layer

World Wide Web:

- From 1990s and 2000s, web sites and web contents grew exponentially.
  - ✓ Sites :
    - Amazon [1994, market value $50b]
    - eBay [1995, market value $30b]
    - Google [1998, mark value $150b]
    - Facebook [2004, market value $15b]

- New ideas are still striking and making web site reach.



Presentation & Application Layer

- **World Wide Web**:
  - Many of the new ideas come from the students
    - ✓ For example:
      - Mark Zuckerberg was a Harvard student when he started Facebook
      - Sergey Brin and Larry Page were standford students when they started Google.
  - Perhaps you will come up with the next big THING.



Presentation & Application Layer

- **World Wide Web**:
  - In 1994, CERN and M.I.T. signed an agreement setting up theW3C [World Wide Web Consortium]
    - It is the organization devoted to further developing the web, standardizing protocols, and encouraging interoperability between sites.



Presentation & Application Layer

- **Multimedia**:
  - The idea of sending audio and video over the Internet has been around since the 1970s
    - ✓ It is only around 2000 that real-time audio and real-time video traffic has grown
  - Example
    - ✓ Live streaming of Audio and Video
      - Audio compression algorithm [MP3, MP4]
      - Video compression [MPEG, JPEG]



Presentation & Application Layer

Internet Protocol Security (IPSec)

- The Internet Protocol is the main routing protocol used on the Internet
  - It designates where data will go using IP addresses.
- IPSec is secure because it adds encryption and authentication to this process.
- IPSec is the Network Layer protocols
  - It is a group of protocols for securing connections between devices.
  - It helps keep data sent over public networks secure.
  - It is often used to set up VPNs, and it works by encrypting IP packets, along with authenticating the source where the packets come from.



Internet Protocol Security (IPsec)

- IPSec protects data as it travels between devices over untrusted networks like the Internet.
- Modes of Operation:
  1. Transport Mode
  2. Tunnel Mode

Transport Mode

- Only the payload (data) is encrypted and authenticated.
- Commonly used for end-to-end communication between two devices.
- Example: Securing communication between two servers.



Internet Protocol Security (IPsec)

- IPSec protects data as it travels between devices over untrusted networks like the Internet.
- Modes of Operation:
  1. Transport Mode
  2. Tunnel Mode

Tunnel Mode

- The entire IP packet (including the header) is encrypted and encapsulated in a new packet.
- Used for VPN connections, where data from multiple devices is encrypted over a public network.



Internet Protocol Security (IPSec)

- IPsec is not one protocol, but a suite of protocols.
- The following protocols make up the IPsec suite:
  1. Authentication Header (AH):
  2. Encapsulating Security Protocol (ESP):
  3. Security Association (SA):



Presentation & Application Layer

Internet Protocol Security (IPSec)

- IPsec suite:
  1. Authentication Header
  2. Encapsulating Security Protocol
  3. Security Association

Authentication Header

- Provides integrity and authentication but no encryption.
- Ensures that data has not been tampered with during transmission.

Integrity

- Receiver receives exactly what it sends by sender
- ✓ **Example:** Monetary transaction

Authentication

- **Receiver needs to be sure of sender's identity**



Internet Protocol Security (IPSec)

- IPsec suite:
  1. Authentication Header
  2. Encapsulating Security Protocol
  3. Security Association

Encapsulating Security Protocol

- Provides encryption, integrity, and authentication.
- Ensures that data remains confidential and protected from unauthorized changes.



Internet Protocol Security (IPSec)

- IPsec suite:
  1. Authentication Header
  2. Encapsulating Security Protocol
  3. Security Association

Security Association

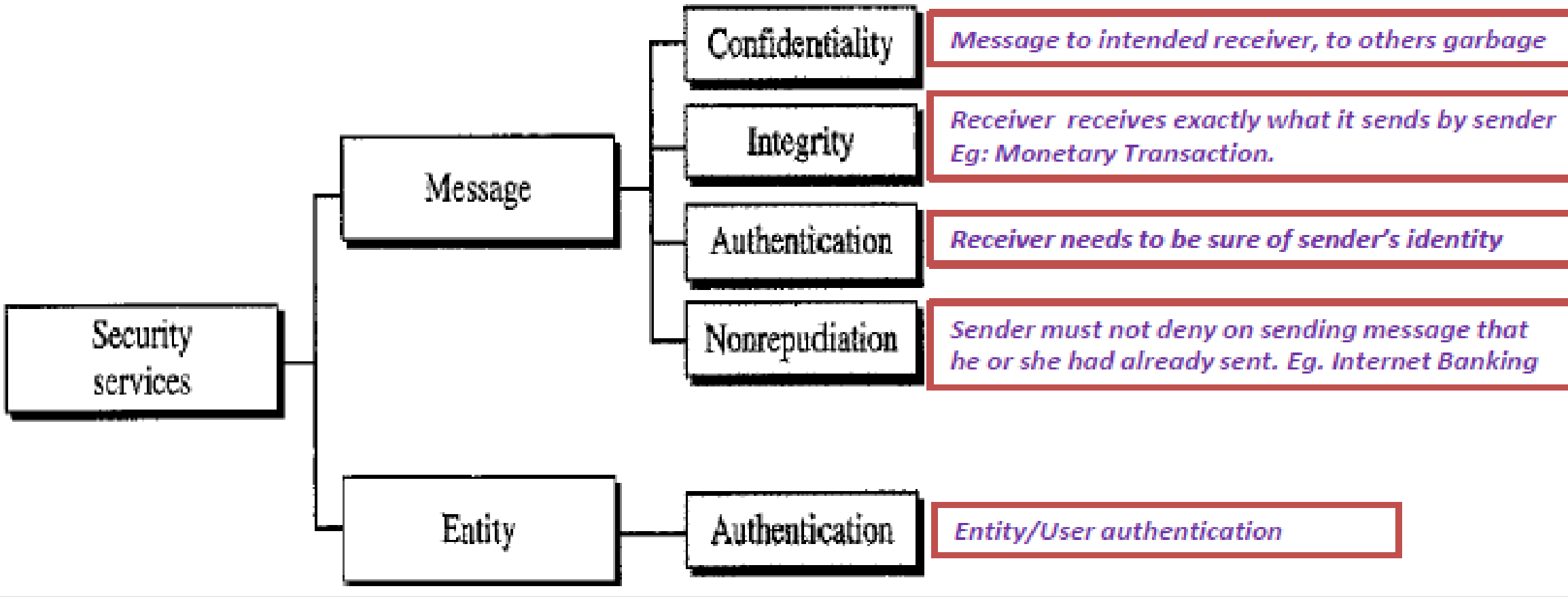
- SA refers to a number of protocols used for negotiating encryption keys and algorithms.
- One of the most common SA protocols is Internet Key Exchange (IKE)..



Presentation & Application Layer

❖ Security Services:

- ➢ Security Services related to Message and Entity





Presentation & Application Layer

Firewalls

- A firewall is a **security device** or software that monitors and controls network traffic based on predetermined security rules.
  - It acts as a barrier between trusted and untrusted networks.
  - It protects and controls the interfaces between a **private network** and a **public network**.

- Types of Firewall:
  1. Packet-Filtering Firewall
  2. Stateful Inspection Firewall
  3. Application Firewall
  4. Next-Generation Firewall



Firewalls

- Types of Firewall:
  1. Packet-Filtering Firewall
  2. Stateful Inspection Firewall
  3. Application Firewall
  4. Next-Generation Firewall

Packet-Filtering Firewall

- Examines headers (source IP, destination IP, port) and allows or blocks packets based on rules.
- Simple but lacks visibility into the packet's contents.

Stateful Inspection Firewall

- Monitors the state of active connections and makes decisions based on the context of the traffic.
- ✓ **Example**: Whether it is part of an established session



Presentation & Application Layer

Firewalls

- Types of Firewall:
  1. Packet-Filtering Firewall
  2. Stateful Inspection Firewall
  3. Application Firewall
  4. Next-Generation Firewall

Application Firewall

- Controls traffic for specific applications
- ✓ **Example:** web, email
  → Blocking traffic to specific websites
  → Filtering email attachments.

Next-Generation Firewall

- Combines traditional firewall features with intrusion detection and application awareness.
- Can perform deep packet inspection to detect malware or attacks.



Virtual Private Network (VPN)

- VPN is an encrypted connection between two or more computers.
- VPN connections take place over **public networks**, but the data exchanged over the VPN is still private because it is encrypted.

- Types of VPN:
  1. Remote Access VPN
  2. Site-to-Site VPN

Remote Access VPN
- Allows users to connect to a private network from remote locations.
- ✓ **Example:** Employees accessing office networks from home.

Site-to-Site VPN
- Connects two or more networks securely over the Internet.
- ✓ **Example:** Secure connection between headquarters and branch offices



Presentation & Application Layer

Wireless Security

- Wireless networks are **prone** to eavesdropping and attacks since data is transmitted over radio waves, making robust security essential.
- Wireless networks are vulnerable due to the open nature of radio waves.
  - Without strong security measures, attackers can intercept or manipulate data, gain unauthorized access, and disrupt services.



Presentation & Application Layer

Wireless Security

- Types of wireless threats

  Direction:

  - ✓ Self Study.
  - ✓ Note (Threats.pdf) is uploaded on VLE
    - Prepare 15 questions and answer

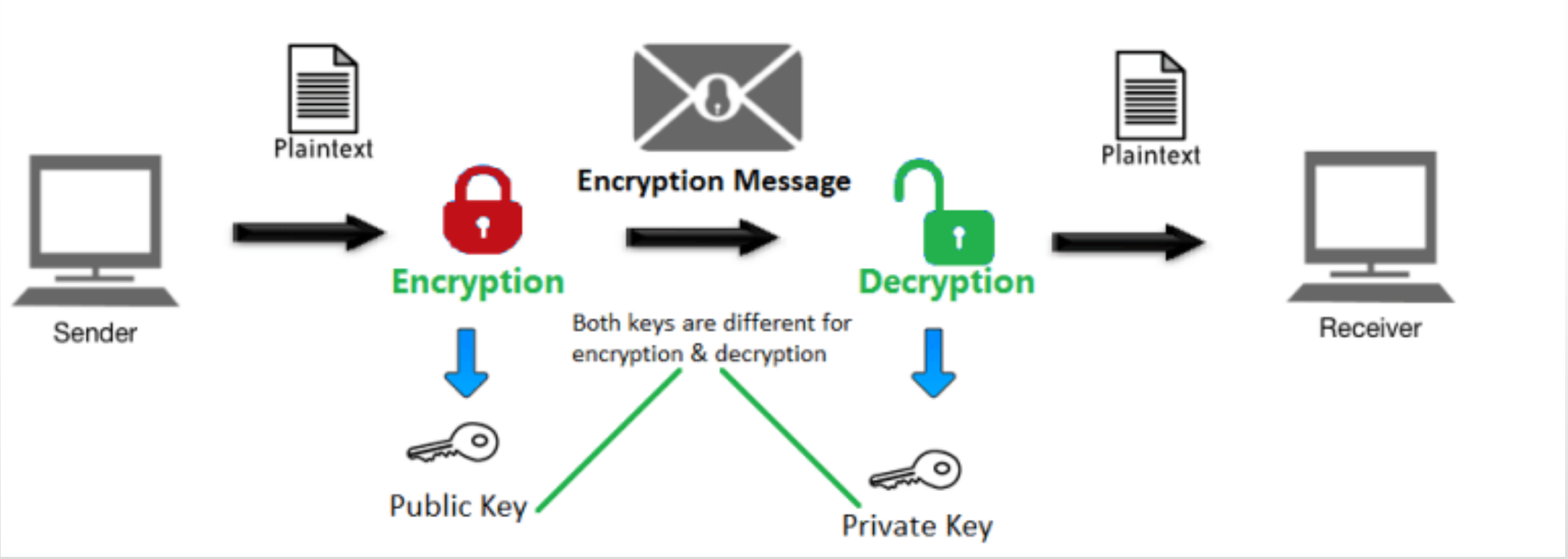


Shared Key Authentication (SKA)

- Authentication using a **shared secret key** relies on both the client and server having a **pre-shared key** (PSK) known only to them.
  - This ensures that both parties can verify each other's identity before communication.
  - It is widely used in **wireless networks**, virtual private networks (VPNs), and network services that require lightweight and fast authentication mechanisms.

Shared Key Authentication (SKA)

- How it works?





Shared Key Authentication (SKA)

- The following are the shared secret key authentication protocols:

1. Wi-Fi Protected Access – Pre-Shared Key (WPA-PSK)
2. IPsec with PSK
3. RADIUS and Shared Key Authentication



Presentation & Application Layer

Wi-Fi Protected Access-Pre-Shared Key (WPA-PSK)

- Commonly used in home and small office wireless networks.
  - How it works:
    - Users enter a shared password (PSK) on their devices to connect to the wireless network.
    - Both the device and the router use the same key to encrypt and authenticate communication.
  - Vulnerabilities:
    - If the password is weak or shared publicly, the network is vulnerable to brute force attacks.
    - Regularly changing the PSK and using complex passwords mitigates the risk.



IPSec with PSK

- In Virtual Private Networks (VPNs), IPsec uses a pre-shared key to authenticate endpoints.
  - How it works:
    - Both the client and the server configure the same PSK before establishing the VPN connection.
    - During connection setup, the PSK is used to mutually authenticate both parties.
    - After authentication, IPsec establishes a secure tunnel for encrypted communication.
  - Challenges:
    - Secure distribution of the PSK to all endpoints.
    - Requires careful management of keys to avoid unauthorized access.



Presentation & Application Layer

RADIUS and Shared Key Authentication

- RADIUS (Remote Authentication Dial-In User Service) is often used in enterprise networks to authenticate users and devices.
  - How it works:
    - Both the RADIUS server and the client device (such as a Wi-Fi router or switch) share the same secret key.
    - When a device or user tries to authenticate, the RADIUS server uses the shared key to verify the request.
  - Challenges:
    - If the shared key is compromised, all devices using that key are at risk.
    - It’s essential to rotate the shared key periodically.



Presentation & Application Layer

Advantages of Shared Secret Key Authentication

1. **Simplicity:** Easy to implement and understand, making it suitable for small networks and systems.
2. **Low Overhead:** Lightweight compared to public key infrastructure (PKI), making it ideal for IoT devices and wireless networks.
3. **Speed:** Faster than other methods such as certificate-based authentication since it involves fewer computational steps.



Presentation & Application Layer

Disadvantages of Shared Secret Key Authentication

1. Key Distribution Problem
2. Scalability Issues
3. Compromised Keys
4. Lack of Forward Secrecy

Key Distribution Problem

- Securely distributing the shared key to all participants can be difficult, especially in large or decentralized networks.
- If the key is transmitted insecurely, it can be intercepted and used by attackers.



Presentation & Application Layer

Disadvantages of Shared Secret Key Authentication

1. Key Distribution Problem
2. Scalability Issues
3. Compromised Keys
4. Lack of Forward Secrecy

Scalability Issues

- ❑ Managing multiple shared keys becomes complex as the number of users or devices increases.
- ✓ For n devices, you need n(n-1)/2 keys if each device has a unique key for every other device.

- n is the total number of devices.
- Each device needs a unique key for secure communication with every other device.
- Since two devices share the same key, we avoid double-counting (e.g., Device A to B is the same as B to A).



Presentation & Application Layer

Scalability Issues

Example 1: 3 Devices (A, B, C)

- n = 3, so the number of keys is:

$$\frac{3(3-1)}{2} = \frac{6}{2} = 3 \text{ keys}$$

Required Keys:

- A ↔ B: **Key1**
- A ↔ C: **Key2**
- B ↔ C: **Key3**

Example 2: 4 Devices (A, B, C, D)

- n = 4, so the number of keys is:

$$\frac{4(4-1)}{2} = \frac{12}{2} = 6 \text{ keys}$$

Required Keys:

- A ↔ B: **Key1**
- A ↔ C: **Key2**
- A ↔ D: **Key3**
- B ↔ C: **Key4**
- B ↔ D: **Key5**
- C ↔ D: **Key6**

Example 3: 5 Devices (A, B, C, D, E)

- n = 5, so the number of keys is:

$$\frac{5(5-1)}{2} = \frac{20}{2} = 10 \text{ keys}$$

Required Keys:

- A ↔ B: **Key1**
- A ↔ C: **Key2**
- A ↔ D: **Key3**
- A ↔ E: **Key4**
- B ↔ C: **Key5**
- B ↔ D: **Key6**
- B ↔ E: **Key7**
- C ↔ D: **Key8**
- C ↔ E: **Key9**
- D ↔ E: **Key10**



Presentation & Application Layer

Disadvantages of Shared Secret Key Authentication

1. Key Distribution Problem
2. Scalability Issues
3. Compromised Keys
4. Lack of Forward Secrecy

Compromised Keys

- If an attacker obtains the shared key, they can impersonate any participant and decrypt the communication.
- There is no way to revoke or change the key without updating it across all devices.

Lack of Forward Secrecy

- Shared key systems do not typically offer forward secrecy, meaning if a key is compromised, all past communications encrypted with that key can be decrypted.



Presentation & Application Layer

Security Risks and Mitigations

- Brute-force Attack
- Man-in-the-Middle Attack
- Key Leakage

Brute-Force Attack

- Attackers try all possible keys until they find the correct one.
- Mitigation:
  - ✓ Use long, complex keys that are resistant to brute force attacks.

Man-in-the-Middle Attack

- An attacker intercepts communication between two parties and uses the shared key to decrypt it.
- Mitigation:
  - ✓ Use session keys that are exchanged after the initial authentication for added security.
  - ✓ Implement **certificate-based encryption** where possible.



Presentation & Application Layer

Security Risks and Mitigations

- Brute-force Attack
- Man-in-the-Middle Attack
- Key Leakage

Key Leakage

- If the shared key is stored improperly, it can be leaked or stolen.

- Mitigation:
  - ✓ Store keys in secure hardware modules.
  - ✓ Rotate keys regularly and limit their lifespan.



References

References:

1. Stallings, W., Data & Computer Communication, Prentice-Hall
2. Tanenbaum, Computers Networks, PHI
3. Forouzan, B.A., Data Communication & Networkings, 4th Edition.
4. Mr. Yeshi Wangchuk's CCN Class Slides (2014), College of Science and Technology, Rinchending, Phuentsholing, Bhutan



See you in next class.